

Semantic Bridge Expansion : IBM 3090 Channel Subsystem (CSS)

Overview

This document outlines the architectural expansion of the SymbioStreams Semantic Bridge to support the legacy IBM 3090 Channel Subsystem (CSS). The bridge will enable modern machine-readable assets to interface with mainframe I/O operations through standardized data mapping.

1. CSS Architectural Components

The expansion focuses on the following primary CSS components:

- Subchannel: The logical representation of an I/O device within the CSS.
- Channel Path Identifier (CHPID): The physical link between the CSS and the control unit.
- Channel Command Word (CCW): The basic unit of control for the mainframe I/O.
- Operation Request Block (ORB): The control block used by the CPU to initiate I/O operations.

2. Mapping Strategy

Modern JSON/YAML metadata from SymbioStreams will be mapped to traditional IBM EBCDIC-encoded control blocks:

- Device Metadata -> Subchannel Information Block (SCHIB)
- Protocol Definitions -> CCW Chaining Logic
- Hardware Telemetry -> Interruption Response Block (IRB)

3. Implementation Plan

- Schema Definition: Create a JSON-to-Mainframe schema mapping for CCW generation.
- Translation Layer: Implement an EBCDIC-UTF8 bridge for semantic data parity.
- Validation: Simulate a START SUBCHANNEL (SSCH) instruction using the mapped metadata.

4. Next Steps

- Verify the specific subchannel ranges for the targeted IBM 3090 environment.
- Finalize the CCW command set for the SymbioStreams hardware assets.